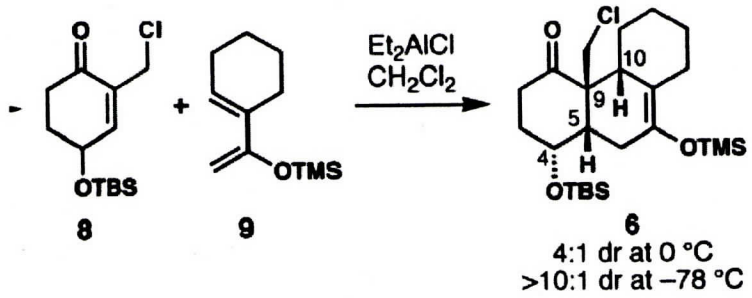
Organic Cumulative Exam April 4, 2014

All questions are based on the article:

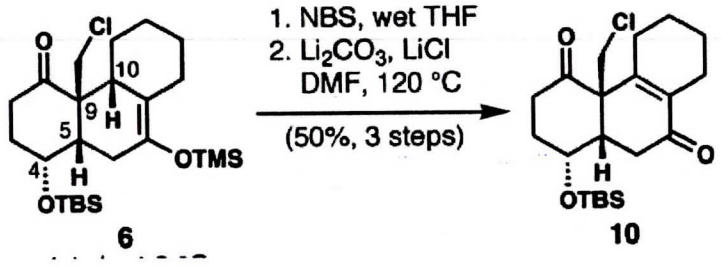
A Synthesis of the ABC Tricyclic Core of the Clionastatins Serves To Corroborate Their Proposed Structures Samuel S. Tartakoff and Christopher D. Vanderwal
Org. Lett., 2014, 16 (5), pp 1458–1461. **DOI:** 10.1021/ol500265v

1. Rationalize the *regioselectivity* of the reaction below.

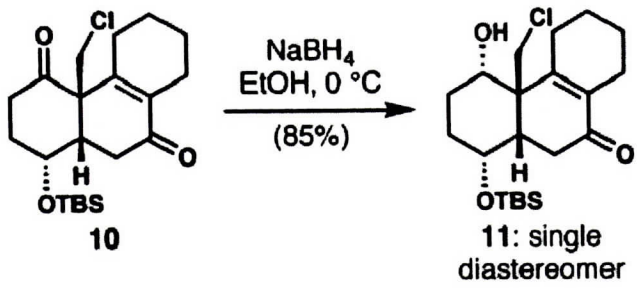
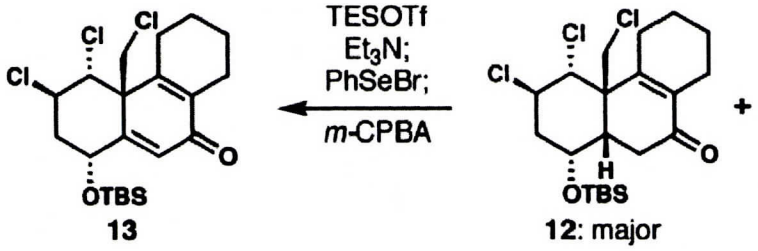


2. Draw the transition state of the reaction above that explains the *stereochemical* outcome. Is the transition state sterically or electronically controlled? Explain your answer.

3. Draw the structure of the intermediate after step 1 (NBS, wet THF).



4. Rationalize the *chemoselectivity* of reduction of the isolated ketone versus the enone in the following reaction.



5. Draw all of the intermediates and the *mechanism of the final step* leading to compound **13** in the following reaction.